HYEMIN KIM

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RESEARCH INTERESTS

Trustworthy & Human-Centric AI, Explainable AI, Safe Reinforcement Learning, Controllable Autonomous Systems.

EDUCATION

Master of Science in Computer and Information Technology August 2019 – December 2021
Purdue University, West Lafayette, IN, USA

Bachelor of Science in Convergence Software March 2014 – February 2019
Sogang University, Seoul, South Korea

RESEARCH & ENGINEERING EXPERIENCE

Software Engineer Jul 2023 – Jun 2026
Applied Intuition (formerly EpiSci) Seoul, South Korea

- **Aerial Combat Multi-Agent RL Platform (Axion-RL)**
  - Engineered a distributed multi-agent RL platform in Go and Python (Ray/RLlib, PyTorch) to train and evaluate high-speed flight autonomy policies.
  - Implemented an Adversarial League framework (PBT + OpenSkill) to auto-curate competitive self-play curricula and continuously evaluate policy robustness against adversarial opponents.
  - Migrated legacy training pipelines to a modular, multi-fidelity simulation architecture (Point-Mass to JSBSim 6DOF), enabling rapid policy evaluation across varying environmental abstractions.
  - Implemented gRPC-Web service boundaries and low-latency IPC to isolate Ray Python worker execution from Go platform runtimes, supporting real-time MCAP flight telemetry streams.
- **GEO-LOCK: Vision-Based Positioning System in GPS-Denied Environments**
  - Formulated a vision-based horizontal localization pipeline for UAVs using satellite reference maps and query aerial imagery targeting GPS-denied environments.
  - Fine-tuned SegFormer (MIT-B0) models on multi-city satellite datasets (mIoU 0.74) and developed a runtime scaling mechanism leveraging camera geometry and Ground Sample Distance (GSD) alignment.
  - Implemented a multi-stage template matching engine combining global IoU search, local ROI scanning, and 3-point parabolic sub-pixel refinement, achieving a mean Absolute Trajectory Error (ATE) of 2.31m.
  - Built a web-based real-time simulation visualizer using Cesium and Google Photorealistic 3D Tiles to evaluate 2D ATE metrics and validate baseline pipeline throughput.
- **SUAS Tactical AI: Real-Time Drone Swarm Autonomy Infrastructure**
  - Engineered a ROS2-based middleware interface via ZeroMQ and Socket.IO, connecting web ground control units with PX4 flight controllers for real-time telemetry streaming and per-drone command execution.
  - Containerized multi-agent execution environments using Docker and FastDDS to support seamless transition between simulated (Gazebo) and live flight hardware (VOXL PX4).
  - Developed a Node.js/JavaScript GCS interface integrating RTSP video streaming and mission planning tools, contributing to a successful system demonstration for U.S. military stakeholders.

- **Gamebreaker: High-Throughput Analytics Pipeline & Web Application**
  - Designed an automated data ingestion pipeline using Python and Riot API to process large-scale historical match data, extract feature matrices for tabular modeling, and evaluate match prediction baselines.
  - Provisioned scalable AWS cloud infrastructure using Terraform and configured automated CircleCI deployment workflows, ensuring reliable continuous integration and delivery.
  - Developed and optimized a web platform initially in React, transitioning to Next.js for server-side rendering (SSR) to improve page rendering speed and data-fetching efficiency.

Software Engineer Jun 2022 – Mar 2023
Koh Young America San Diego, CA, USA

- Built automated CI/CD pipelines using Jenkins, Python, and CMake to streamline build tasks, static analysis, unit testing, and SBOM generation across Windows and Linux environments for C++/C# applications.
- Converted legacy Visual Studio C++ projects to CMake for dynamic library resolution, established an internal package management server, and provisioned isolated VM test environments to automate cross-platform deployment workflows.

TEACHING EXPERIENCE

Graduate Teaching Assistant, Purdue University Fall 2019, Spring 2020, Spring 2021
Course: CNIT 31500 – Systems Programming

- Mentored 100+ undergraduate students in lower-level C/C++ systems programming labs, guiding hands-on implementations of algorithms, data structures, and system concepts on UNIX/Linux platforms.
- Designed automated unit test cases, evaluated lab assignments and examinations, and held weekly office hours to assist with code debugging and system concepts.

TECHNICAL SKILLS

- **Languages:** Python, TypeScript, Go, Bash, C/C++
- **AI & Reinforcement Learning:** PyTorch, Ray/RLlib, Gymnasium, Hugging Face, OpenCV
- **Robotics & Simulation:** ROS2, PX4, Gazebo, FastDDS, JSBSim
- **Infrastructure & MLOps:** Docker, AWS, Terraform, CI/CD (Jenkins, CircleCI), gRPC, CMake, Linux, Git